

Foundational Research & Advances in Quantum Alternating Operator Ansatz (QAOA+)

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

Published in: ACM Transactions on Quantum Computing / Physical Review A (2023) | Archival ID: QV-QC-79

1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Alternating Operator Ansatz (QAOA+) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Instead of letting the quantum search wander into illegal moves and punishing it later, QAOA+ designs the quantum engine so it physically CANNOT take an illegal step."

3. Mathematical Formulation & Unitary Rule

$$[U_M(\beta), \mathcal{P}_{\text{feasible}}] = 0, \quad \text{quad } H_{XY} = \sum_{\langle i, j \rangle} J_{ij} \sigma_i^x \sigma_j^x$$

The mixer unitary commutes with the projector onto the feasible subspace, ensuring amplitude remains 100% within valid constraints.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
import numpy as np

beta = np.pi / 4
qc = QuantumCircuit(2)
# Conserves total excitation number: |01> <-> |10>
qc.rxx(beta, 0, 1)
qc.ryy(beta, 0, 1)
print(qc.draw(output='text'))
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Penalty method constraints distort the feasible subspace, making search heuristic infeasible Hilbert space conf

7. Key Applications & Future Research Directions

- Portfolio optimization with strict cardinality constraints
- Traveling Salesperson Problem (TSP) and permutation routing
- Exact k-satisfiability (k-SAT) problem solving

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Optimization pipelines.